

Fig. 4: An actual-size, single-side PCB for the transmitter circuit (Fig. 2)

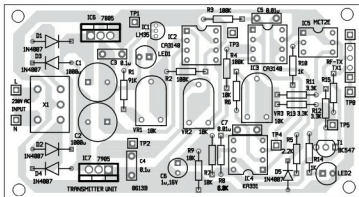


Fig. 5: Component layout for the PCB in Fig. 4

of some components is necessary as follows.

Transmitter unit adjustment

Offset adjustment. Measure the voltage at TP3 with respect to ground by using a digital voltmeter. Disconnect R1 and short the inverting input to ground. Switch on the $\pm 5V$ supply. Adjust preset VR2 to get 0V reading on the voltmeter.

Gain adjustment. Operational amplifier IC2 acts as a unity-gain amplifier. Reconnect resistor R1. Connect pin 3 of IC1 directly to ground by shorting LED1. Keep sensor IC1 at a fixed temperature. If room temperature is constant, keep the sensor free to sense the room temperature. Measure output voltage of the sensor with a digital mV meter. Connect another digital mV meter to read the output voltage at TP3. Adjust preset VR1 to get the same voltage reading on TP3 but with negative polarity. If required, change the value of R1.

V-F gain adjustment. The V-F converter should generate frequency in hertz exactly equal to the voltage input in millivolts. To get this, the gain of the V-F converter is to be adjusted. For this, connect a frequency counter between test point TP5 and ground, and a digital mV meter between TP3 and ground. Adjust VR3 to get a frequency in Hz that is equal to the meter reading in mV. Sensor IC1 must be kept at a fixed temperature.

Receiver unit adjustment

F-V gain adjustment. The F-V converter should generate output voltage in millivolts exactly equal to the input frequency in Hz. To get this, the gain of the F-V converter is adjusted. To do this, connect a frequency counter to pin 3 of IC9 and a digital millivolt meter at pin 1 of F-V converter IC10. Switch on the transmitter and the receiver units. The frequency counter should display the frequency transmitted from the transmitter unit. LED3 connected to output pin 3 of the NAND

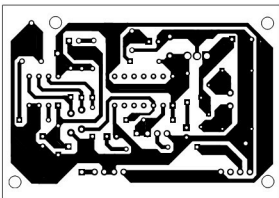


Fig. 6: An actual-size, single-side PCB for the receiver circuit (Fig. 3)

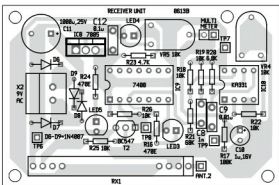


Fig. 7: Component layout for the PCB in Fig. 6

gate indicates the incoming frequency signal. Adjust preset VR4 to get a mV reading on the voltmeter that equals the frequency in Hz displayed on the frequency counter.

Adjustment of reference voltage. As already stated in the transmitter unit description, pin 3 of temperature sensor IC1 is raised above the ground potential by the reference voltage developed across LED1. At the receiver end, this voltage is to be subtracted to get the actual temperature-sense voltage developed by sensor IC1. To do this, connect the positive lead of the millivolt meter to pin 1 of F-V converter IC10 and negative lead of the meter to the rotary arm of preset VR5. Now adjust preset VR5 to get actual temperature on the meter's screen.

In a properly calibrated system, meter reading should increase or decrease @ $10\text{mV}/^{\circ}\text{C}$. Therefore a 0.250V reading on the mV meter indicates 25°C temperature. ■

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